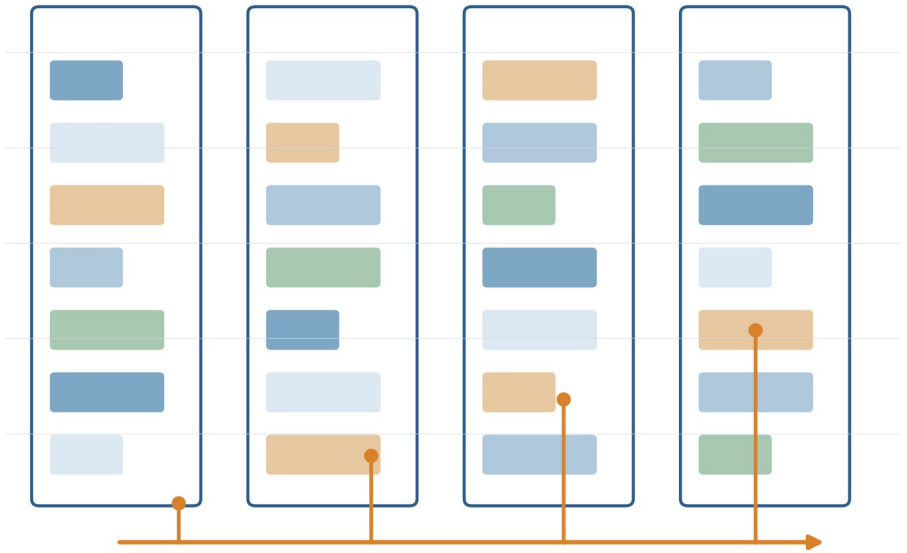
Revisiting Lifetime-Aware VM Allocation

How prediction quality changes server packing

Part 1 The problem
Part 2 What we tested
Part 3 Our improvement

CE7490 Course Project · 26S1 Group 5

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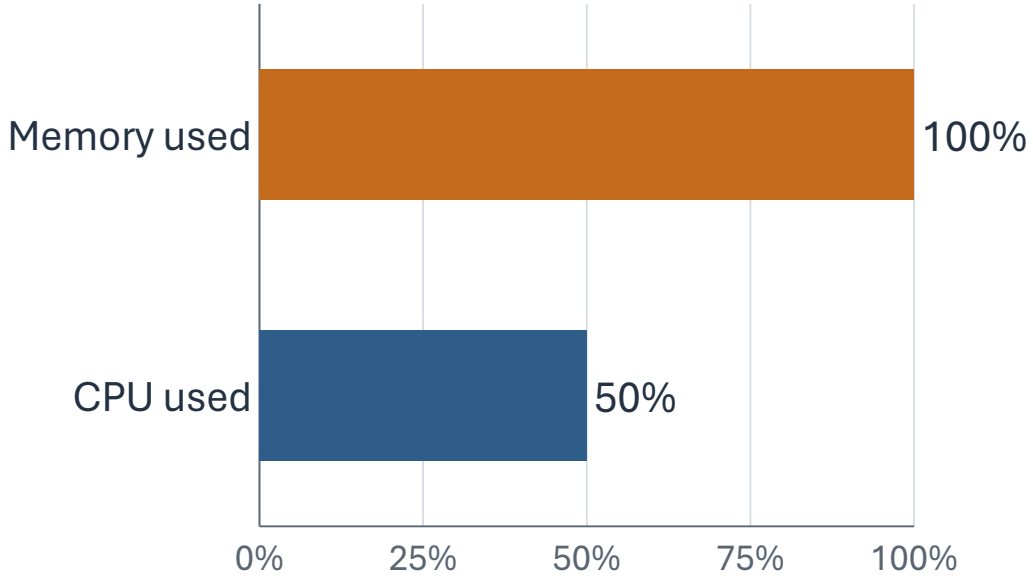
VMs need CPU and memory

VM = a small virtual computer

Host = the physical server
that runs several VMs

A new VM needs both CPU and memory.
Free CPU alone is not enough.

A full-memory example

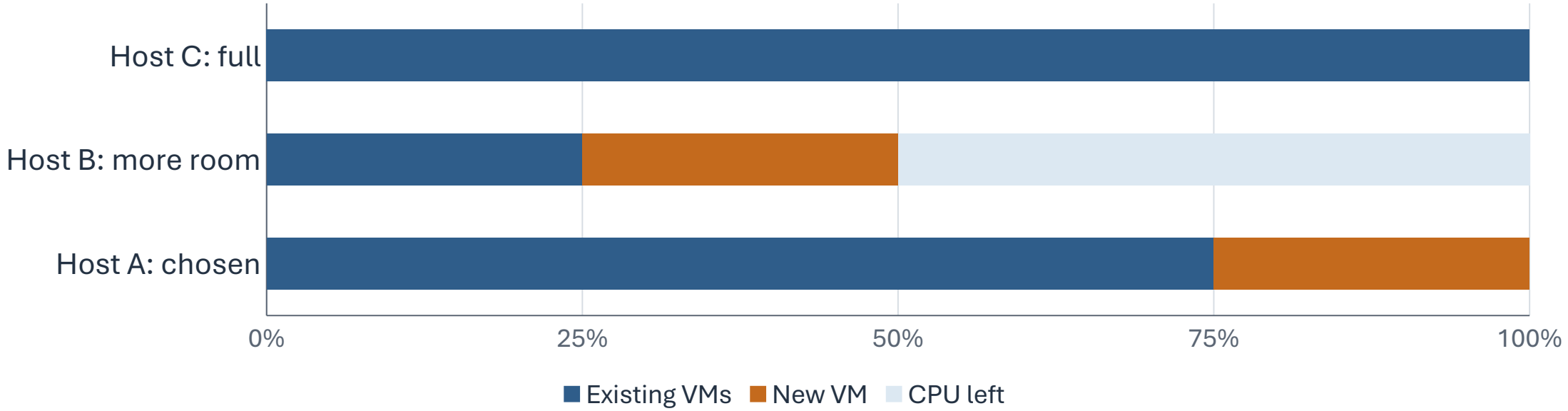


Illustrative capacity values

Best-Fit packs VMs tightly

Best-Fit is our starting point for comparison.

New VM: 25% CPU and 12.5% memory

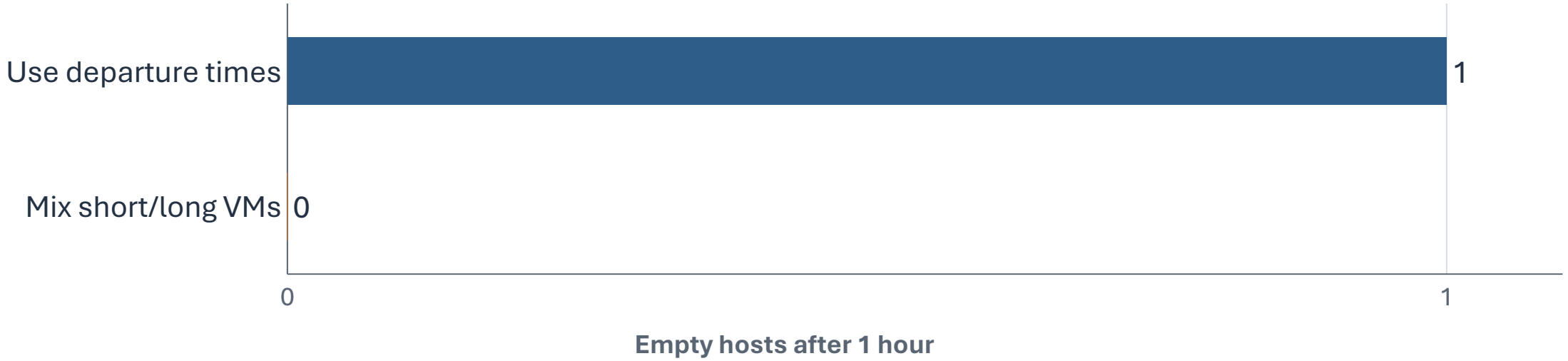


A fits tightly. C has no CPU space left.

LAVA uses expected departure times

LAVA means Lifetime-Aware VM Allocation.

Four VMs: two run for 1 h, two for 10 h. Each host fits two.

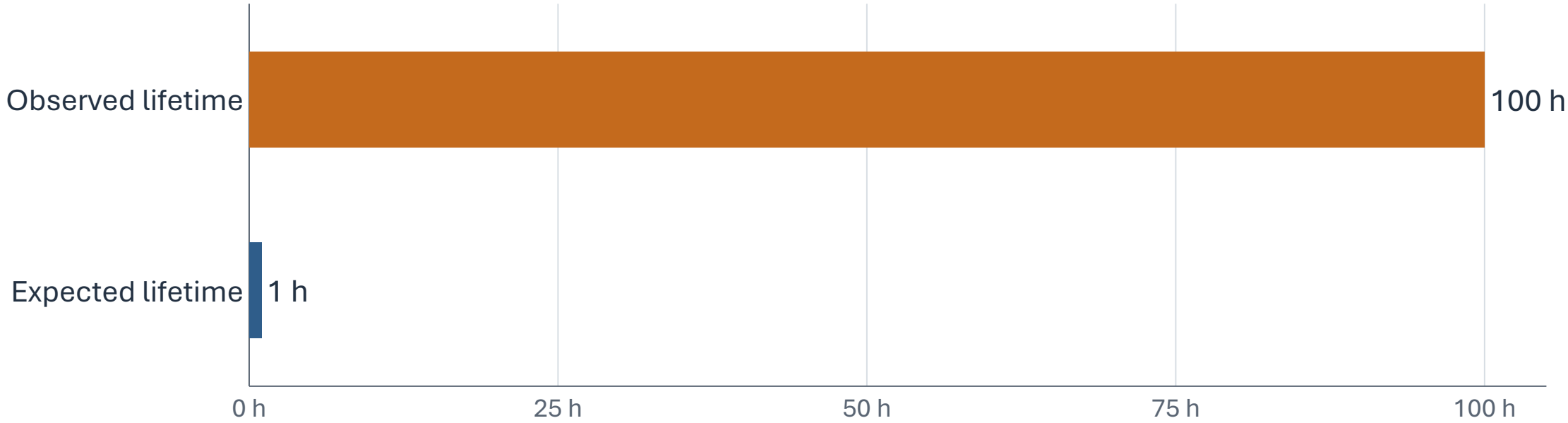


Similar departure times can empty a host earlier.

Illustrative data, not the full LAVA rule. LAVA also uses host states and re-prediction.

Wrong predictions keep hosts busy

One long-running VM can keep the host busy.



The host remains occupied much longer than expected.

Illustrative example, not an experimental trace

Give every method the same workload

Every method gets the same requests and predictions.

Same requests and predictions

- **Best-Fit**
- **LAVA**
- **LAVA with our improvement**

180
physical hosts

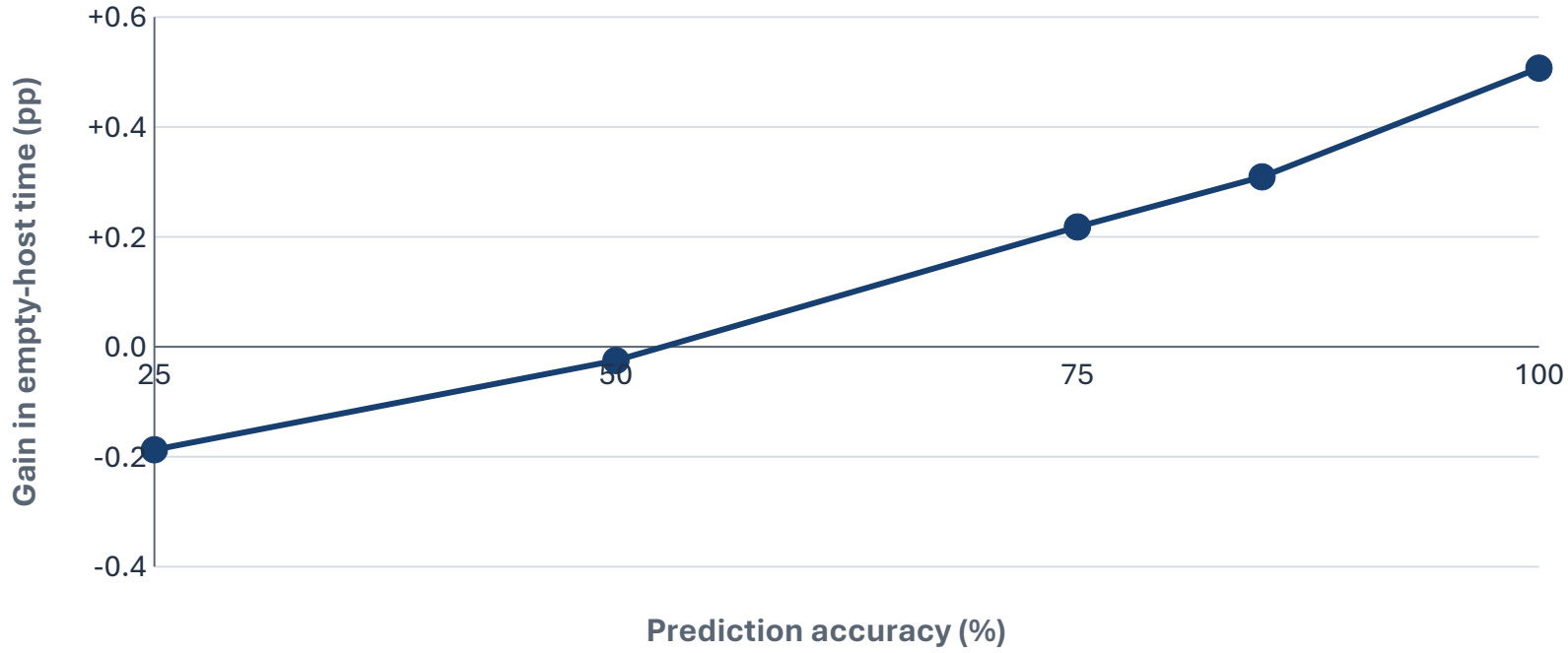
2,500
VMs per workload

100
new test workloads

Goal: keep more hosts empty, while accepting every VM.

Poor predictions can make LAVA worse

Gain over Best-Fit on 100 new test workloads



25% accuracy

-0.187 pp

100% accuracy

+0.507 pp

Accuracy = probability of a near-correct estimate in this controlled test.
pp: 10.0% to 10.2% empty-host time is +0.2 percentage points.

Use confidence to choose a rule

Confidence = probability of the estimated lifetime group.

High confidence: use LAVA. Low confidence: use Best-Fit.

VM-level routing

Each arrival chooses a rule

LAVA or Best-Fit

One shared set of hosts

Host-level routing

Each arrival chooses a pool

LAVA pool or Best-Fit pool

Keep the two modes separate

Confidence routing did not help here

1 **Confidence can be misleading**

A usually short-lived profile gets low confidence.

2 **Almost every request can pass**

Under controlled noise, almost every request passes.

3 **Separate pools waste room**

An empty slot in the other pool may go unused.

Host-CA minus Best-Fit: -0.090 pp (default predictor)

Combine the estimate with usual behavior

Use usual VM behavior to check a new estimate.

Usual behavior	The noisy estimate	Updated estimate
This profile is usually short-lived	300 h	1.52 h

Typical profile mean: 1.07 h

We improve the estimate. We keep the same scheduling rules.

Keep updating while the VM runs

A running VM gives us new information.

The VM is still running.

It has already lasted longer than expected.

Update its predicted time left.

Keep learning while the VM runs.

Better than LAVA in 3 of 4 tests

Our improved estimate, using the same scheduling rules

3 of 4

tests improved over LAVA

100 new test workloads

25%	Better
50%	Better
75%	Not confirmed
85%	Better

At 25%, improvement over Best-Fit remains unconfirmed.

Better estimates seem to matter most

Was it the better estimate or the choice of scheduling rule?

Use the same improved estimate

Compare it with and without switching rules.

Extra effect of switching rules

-0.018 to +0.011 pp

Switching rules added no clear gain here.

Exploratory analysis, not a registered claim

Wrong prior knowledge can hurt

Wrong usual behavior can pull a good estimate in the wrong direction.

Assumed lifetimes cut in half

85% prediction accuracy

-0.116 pp

compared with LAVA

Learn usual behavior from real history, and check for changes.

One exploratory robustness case, 50 test workloads

What should we remember?

Findings

Poor predictions can hurt LAVA.

Routing did not fix the problem here.

Our improvement beat LAVA in 3 of 4 tests.

Limits

Only synthetic workloads tested

Usual behavior and error model assumed known

More real-world tests needed

Questions?

Code, data and report

github.com/xxxmian/
CE7490-P2_Revisiting_LAVA